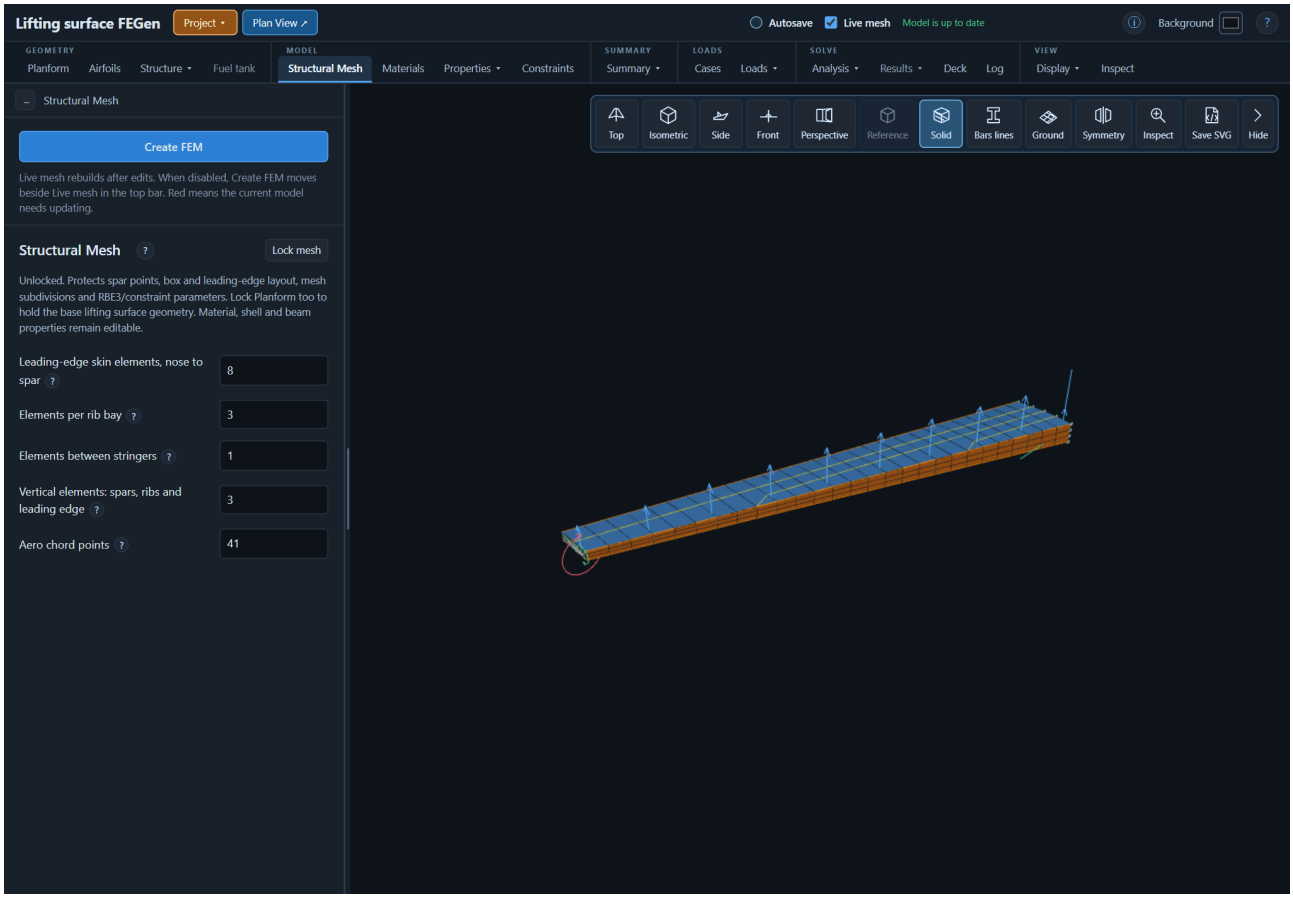
FGen User Guide

Lifting surface FEGen

From a first Study to finite element results



Release guide — 13 October 2026

For the browser application bundled with OpenJFEM. The application, launcher and file formats also use the established name *WingFEGen*.

All illustrated models were created from a fresh, synthetic default lifting surface in an isolated application session. The static-results example is a real SOL101 calculation. The example teaches the workflow; it is not a validated aircraft design.

OpenJFEM project and downloads

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1 Your first result in ten steps

FGen builds a lifting-surface structure, writes a Reference finite element deck and runs JFEM. It can also open an existing Reference deck without reconstructing it as a generated wing. Start with the generated model to learn the controls.

1. Launch the bundled WingFEGen application following the installation guide. Leave its local server running while using the browser.
2. In the startup dialog, choose **Start new study**. Wait until the model is ready. Loading the application alone does not launch an analysis.
3. Open **Project** and save a Study with a descriptive name. This is your editable project, not the solver's results file.
4. Examine **Planform** and **Airfoils**. For this first run, retain the default geometry and structural properties.
5. Open **Structural Mesh**. Use **Create FEM** after an edit when **Live mesh** is off. Check the mesh and the model status before solving.
6. Open **Constraints**. Confirm that the intended root constraint is present. A model that can move freely needs a physically appropriate constraint definition before a static solve.
7. Open **Cases**. Review the enabled load case, load multiplier, load definition and fuel status. Keep one enabled case while learning.
8. Under **Analysis > FE Analysis**, select **101** for a linear static calculation. Review the results folder and solver location if automatic discovery is unavailable.
9. Click **Run in JFEM**. Follow the phase in the bottom-right activity panel. When finished, open **Results > FE Results** and select the case and field.
10. Compare displacement, reactions and stresses with the applied loading. Save the Study again; retain the separate results folder with it.

Use the **?** next to a field for its meaning and units. Hover for quick help; click for the full explanation. Change one group of inputs at a time until the workflow is familiar.

2 Project files, save and autosave

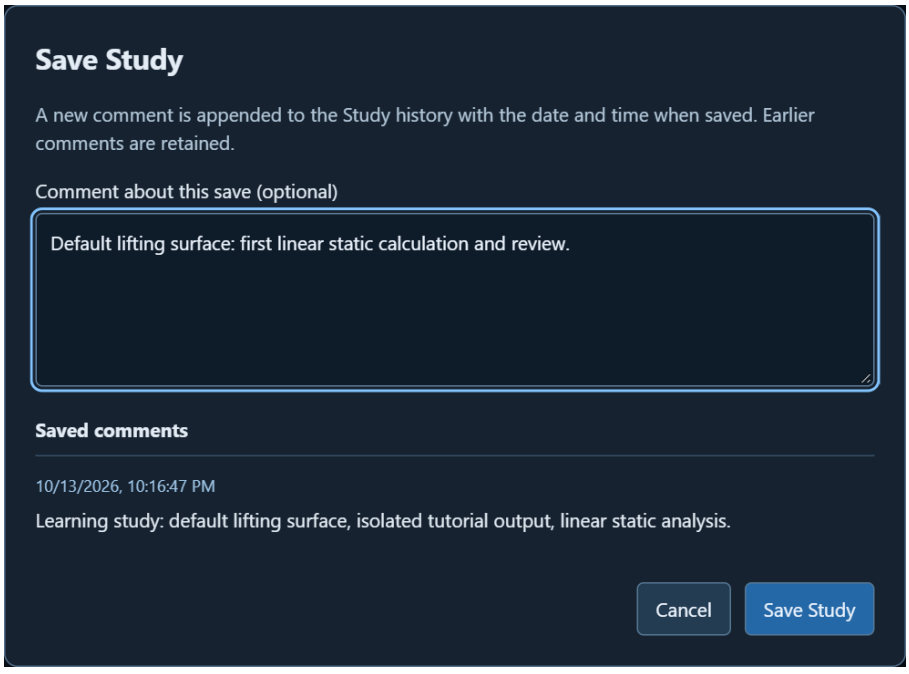
The **Project** menu distinguishes a portable Study, a parameter TOML file and a Reference analysis deck. Choose the format for the next task.

File	Use
Study, `.wingfem.json`	Reopen the editable workspace: model parameters, cases, constraints, embedded reference inputs, drawings, display settings and timestamped comments. Solver results remain separate.
TOML, `.toml`	Exchange generated-model parameter values. Paths can depend on the computer or server working directory. Use a Study for portable embedded geometry and workspace state.
Reference, `.bdf/.dat/.nas`	Inspect or run the FE model and case control in a deck-based workflow. Read it through **Read Reference**.
Geometry/SVG	Share a geometric or visual representation. These are not editable Study backups.

Save Study updates the linked file, choosing a destination on the first save. **Save Study as** chooses another destination. A browser without direct file writing downloads the file instead; check the download destination and keep the intended revision. Add a short comment describing why a change was made. Existing comments remain part of the Study history.

Project and Save Study are orange when the definition has unsaved changes and green when it has been saved. Camera and display changes do not mark the engineering definition dirty, but the current view settings are included in an explicit save.

Autosave, beside **Live mesh**, saves settled definition edits after a Study destination has been selected. It covers changes such as geometry, properties, loads, comments and selected design variables. Camera movement alone does not trigger a save. Keep named revisions for major alternatives; autosave is not a substitute for a revision history or separate results files.



Saving a Study with an explanatory comment. The browser's destination dialog or download follows according to browser support.

The advanced configured-server-file commands are separate from Study saving. Do not use them merely to save a normal browser Study.

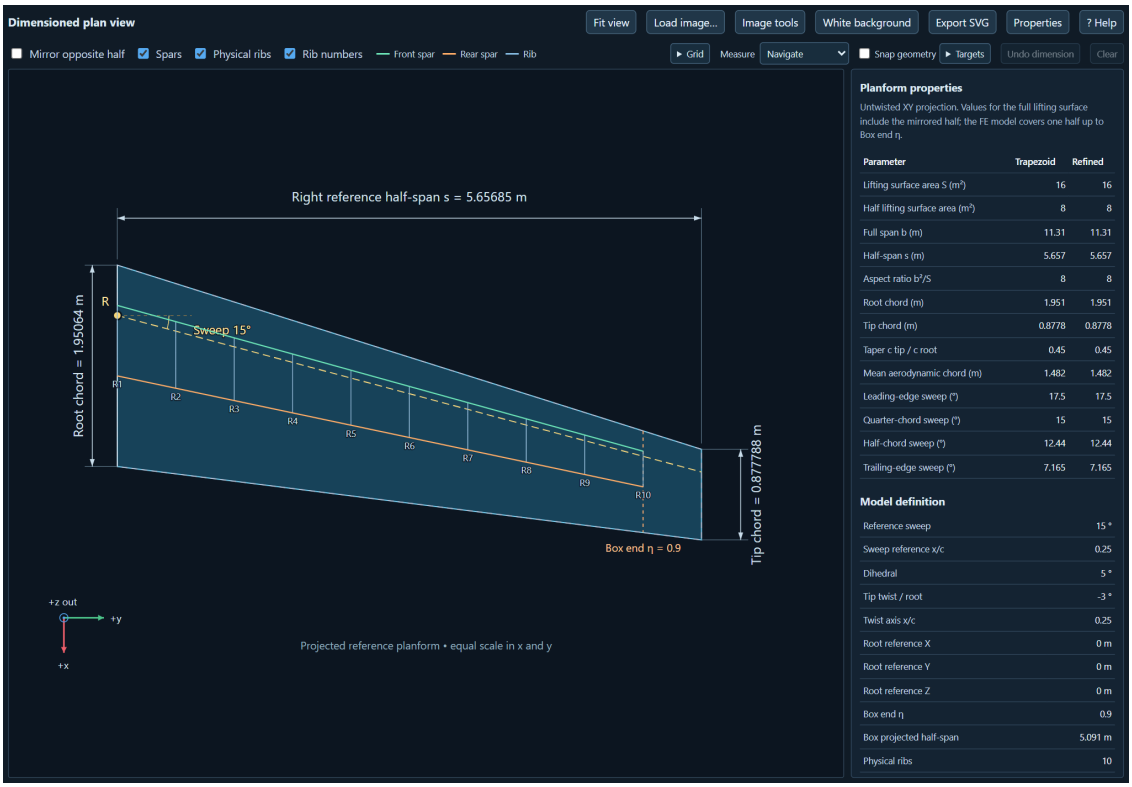
3 Geometry and the plan view

Planform defines the base trapezoid by area, aspect ratio and taper, or by the alternative dimensions offered in the form. Check the drawing and reported span/chords after changing the definition. In the generated model, lengths are metres and angles are degrees unless the field says otherwise. Global $+X$ is aft, $+Y$ runs toward the modeled tip and $+Z$ is upward.

Sweep, dihedral and twist change the three-dimensional surface. Edge and spar control points refine the base planform. Their normalized span position is η, with 0 at the root and 1 at the tip. Retain ordered stations and inspect narrow or sharply varying bays after edits. The displayed base and refined areas explain why a refined outline need not have exactly the base trapezoid's area.

Lock planform protects the planform, aerodynamic edge points and Airfoils from accidental definition changes. Unlock it to edit these inputs. It does not replace saving.

Use the blue **Plan View** button, or **Dimensioned plan view**, to open the synchronized drawing in another tab. Navigate and zoom there, add dimensions and annotations, or place a background image for comparison. Select a measurement mode deliberately; return to **Navigate** to move around the drawing. Snapping helps pick geometric endpoints. Undo or clear dimensions when revising a drawing, and export SVG when a vector drawing is needed.



The separate synchronized plan drawing provides dimensions, annotations, snapping and export controls.

A drawing dimension is an annotation. Edit the corresponding geometry input to change the FE model; moving a camera or annotation does not change physical dimensions. Plan View is unavailable for imported decks without generated lifting-surface metadata.

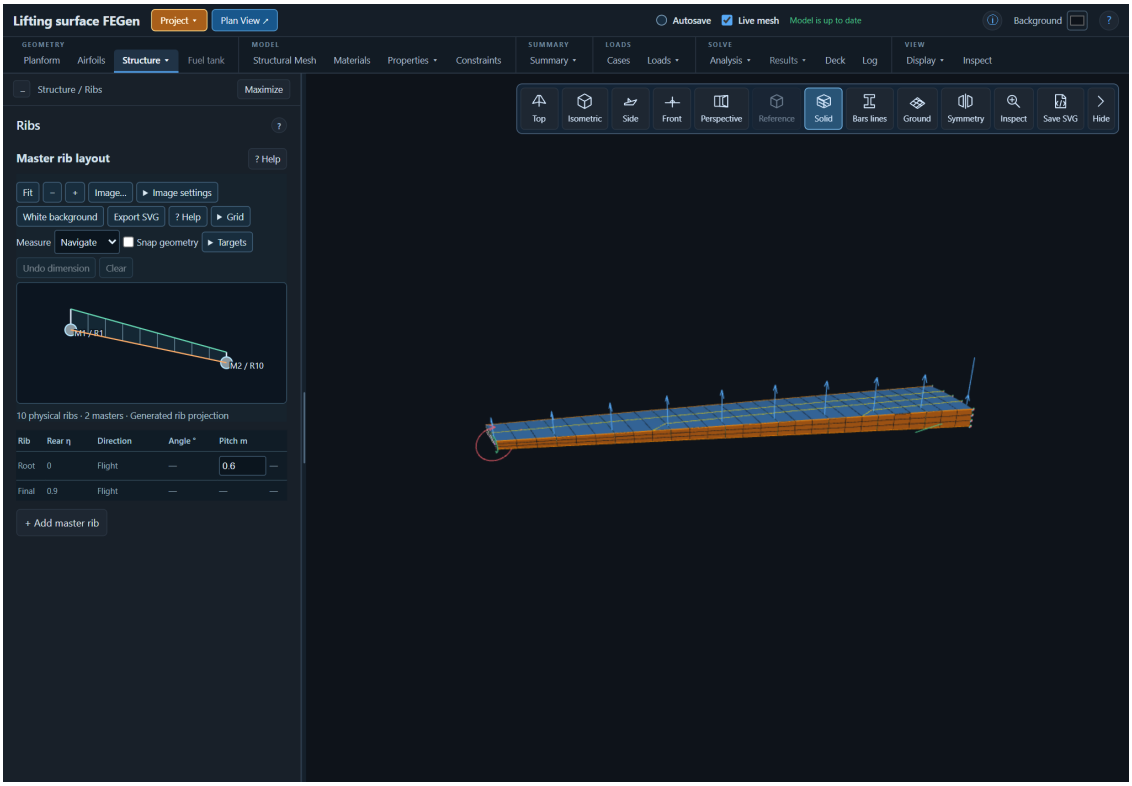
4 Airfoils, spars, ribs and stringers

Airfoils controls the aerodynamic section definition and spanwise interpolation. Read the section preview together with thickness/camber measures. Verify root and tip sections, orientation and transitions before refining the mesh. Surface display sampling and the aerodynamic solver grid are separate from the structural FE mesh.

In **Structure**, define the box limits and the physical rib/stringer layout. Front and rear spar positions determine the load-carrying box. Rib pitch or stations locate physical ribs; changing these changes the structure. Angled ribs should remain consistent with the selected layout and the planform edges.

Structure > Stringers controls pitch, global angle or angles by rear-spar segment, and the normalized **Run-out threshold**. Pitch is a physical spacing, not an FE convergence setting. Inspect stringer terminations and short run-out regions before accepting a layout.

The optional **Leading edge** adds the corresponding structural region. Enable and define it in the geometry controls before assigning its properties. Leading-edge property controls stay disabled when no leading edge is defined. Its skin refinement is in **Structural Mesh**.



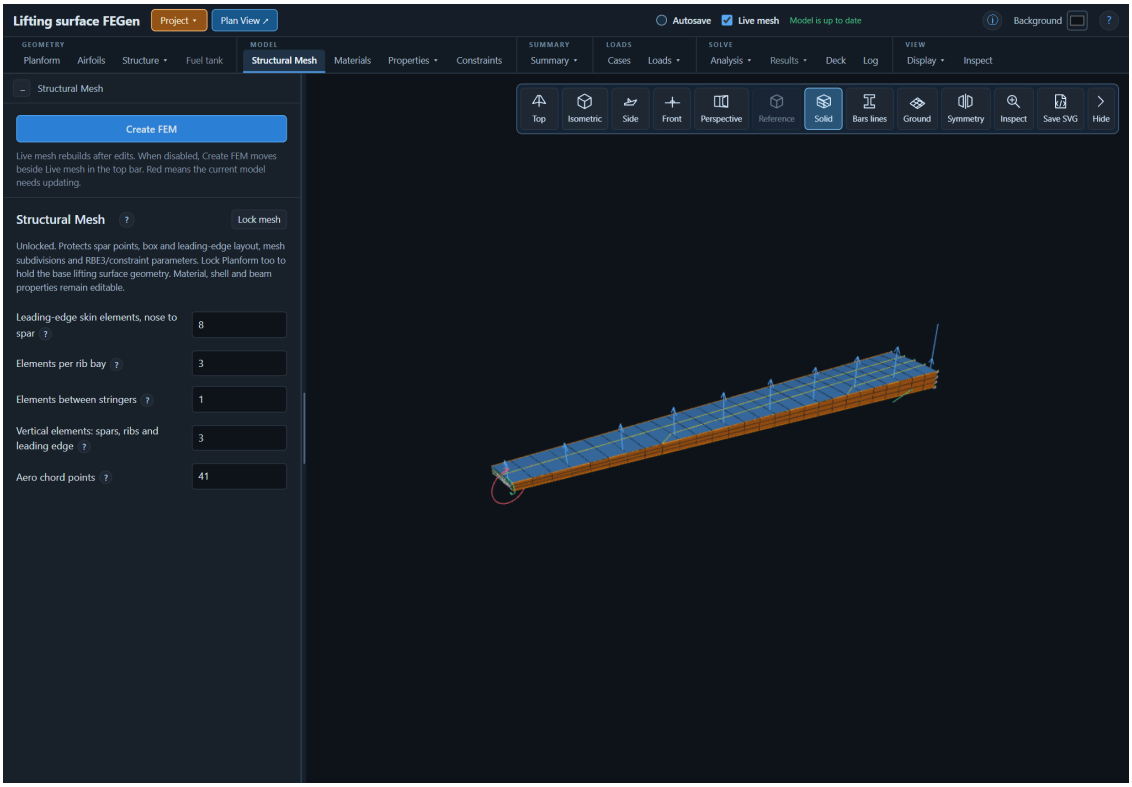
Rib layout is a structural definition. Check the three-dimensional connections as well as the plan drawing.

First choose the physical structure; then refine its mesh. Increasing the number of FE subdivisions does not add a new physical rib or stringer.

5 Create and refine the structural FEM

Create FEM regenerates the current generated model. With **Live mesh** enabled, settled edits trigger regeneration automatically. Disable Live mesh while making several related edits, then rebuild once. The model status must be current before its mesh is used to interpret a new calculation.

Control family	Effect
Structural chord/span subdivisions	Refine FE elements within the physical structural layout. Inspect the resulting node and element counts.
Spar/rib/leading-edge vertical count	A shared count through the structural height. Spars, ribs and the leading-edge connection use matching interface nodes. The default preserves the established mesh.
Leading-edge skin elements	Subdivisions from the nose to the front spar on each upper/lower skin. Refines the optional leading-edge skin independently of the vertical count.
Aero chord points	Samples the displayed aerodynamic surface. It does not replace either structural refinement or VLM discretization.
VLM settings	Define the aerodynamic calculation grid. These belong to Loads, not Structural Mesh.



Structural Mesh and the generated FE structure in isometric view. Mesh controls are separate from display sampling and VLM settings.

A practical convergence check. Save the baseline Study and result. Refine one meaningful FE direction, rerun the same case, and compare a response such as tip displacement or the first relevant buckling factor. Repeat until the response is stable to the accuracy required by the work. Also inspect local distortion, load transfer and property boundaries; a larger count alone is not evidence of an adequate mesh.

6 Materials, properties and constraints

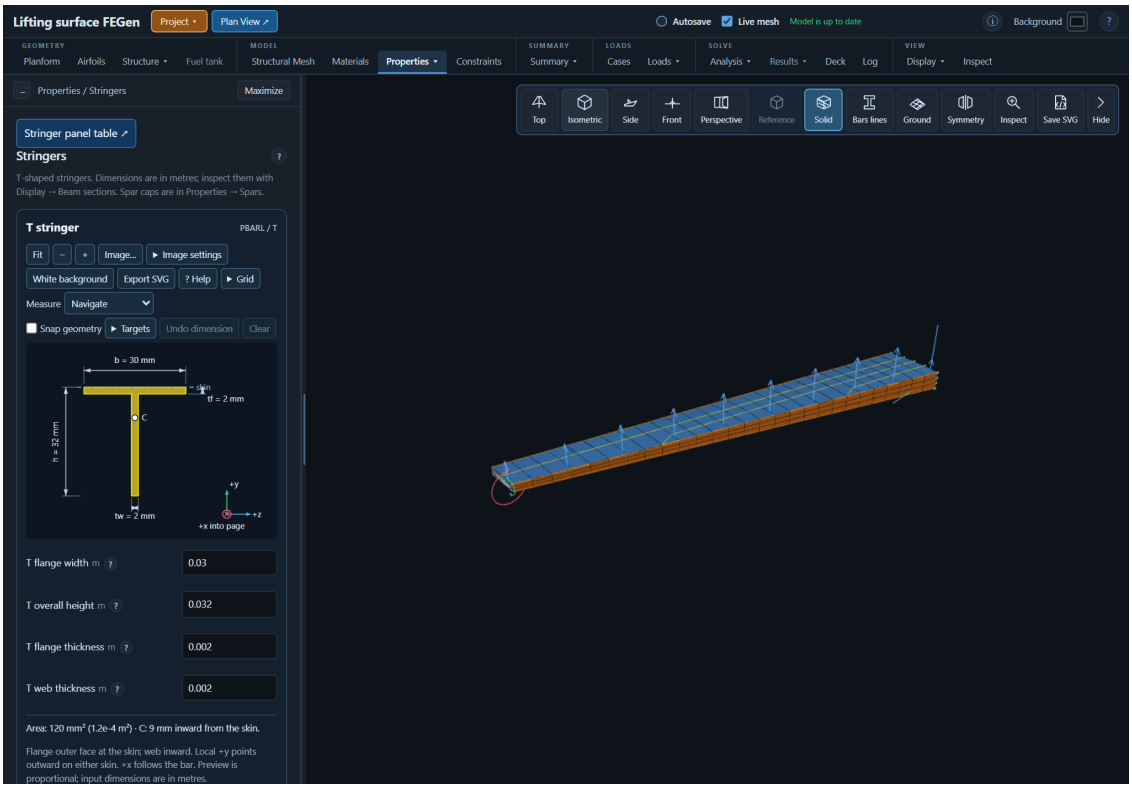
Use **Materials** for the constitutive inputs and density. Check units before entering values: an elastic modulus typed in GPa into a field expecting Pa would change the stiffness by a factor of 10^9. Generated models use consistent N–m input; imported models retain the units of their source deck.

The **Properties** menu separates shell, section, rib, spar and leading-edge definitions. Start with the shared defaults, then use local overrides where a bay or component needs a different property. Review the effective values in the tables, rather than assuming a global change replaced an existing override. Table colors help compare accepted values; they are not acceptance criteria.

Cross-section dimensions affect beam area, bending/torsional stiffness and mass. Skin and web thicknesses affect shell stiffness and mass. Material selection affects both stiffness and density. Check the mass/weight summary after property changes to catch unintended units or scope.

Constraints defines boundary constraints for generated models. Review the constrained nodes and components. In a half-model, a root attachment and a symmetry condition represent different physical assumptions; choose the actual intended boundary condition. Display symbols help verify the location, but hiding a symbol does not remove its constraint.

Display > Axes & labels > Shell material axes shows material directions at shell centers. **Material coordinate labels** can be shown independently. Imported MCID and THETA definitions must be interpreted in their source convention. These directions matter for composites and directional stresses; isotropic in-plane stiffness does not change merely because its reporting axis rotates.



Section-property controls and effective property tables. Review the selected component and any local overrides before editing.

7 Fuel tank and mass accounting

Enable the optional tank, select its geometric bounds and mark dry bays as required. Check the plan representation and the three-dimensional tank. Dry bays are excluded from fuel capacity.

Fill efficiency factor is a uniform usable-volume fraction between 0 and 1; its default is 1. For a non-dry geometric capacity V_g, density ρ and efficiency e,

$$V_{\text{usable}} = eV_g, \qquad m_{\text{full}} = \rho e V_g.$$

The load-case **Fuel status** chooses the percentage of this usable capacity. It is a separate input. For example, a geometric wet-bay volume of $2\,\text{m}^3$ and efficiency 0.9 gives $1.8\,\text{m}^3$ usable full capacity. A 50% case contains $0.9\,\text{m}^3$ under this allowance model.

The allowance is distributed uniformly through the fuel volume. At the same case percentage, efficiency scales fuel mass, inertia and inertial loads but does not change fill height or centroid. It is not a new geometric obstacle, a shifted free surface or a change in structural mass. Use actual tank geometry and dry bays when the spatial distribution itself must change.

Review **Summary** for the structural mass, fuel contribution, center of gravity and inertia. Distinguish gross geometric envelope from usable capacity. A tank can be enabled yet contribute no fuel in an empty case; a disabled tank or a dry bay contributes no fuel. Check both the geometry definition and active case before diagnosing a zero fuel mass.

Suggested check. Compare an empty case and a full case with all other inputs unchanged. Confirm the expected mass change and load change. Then set efficiency to a known fraction and verify the fuel-only contribution scales accordingly. Save the intended factor with the Study.

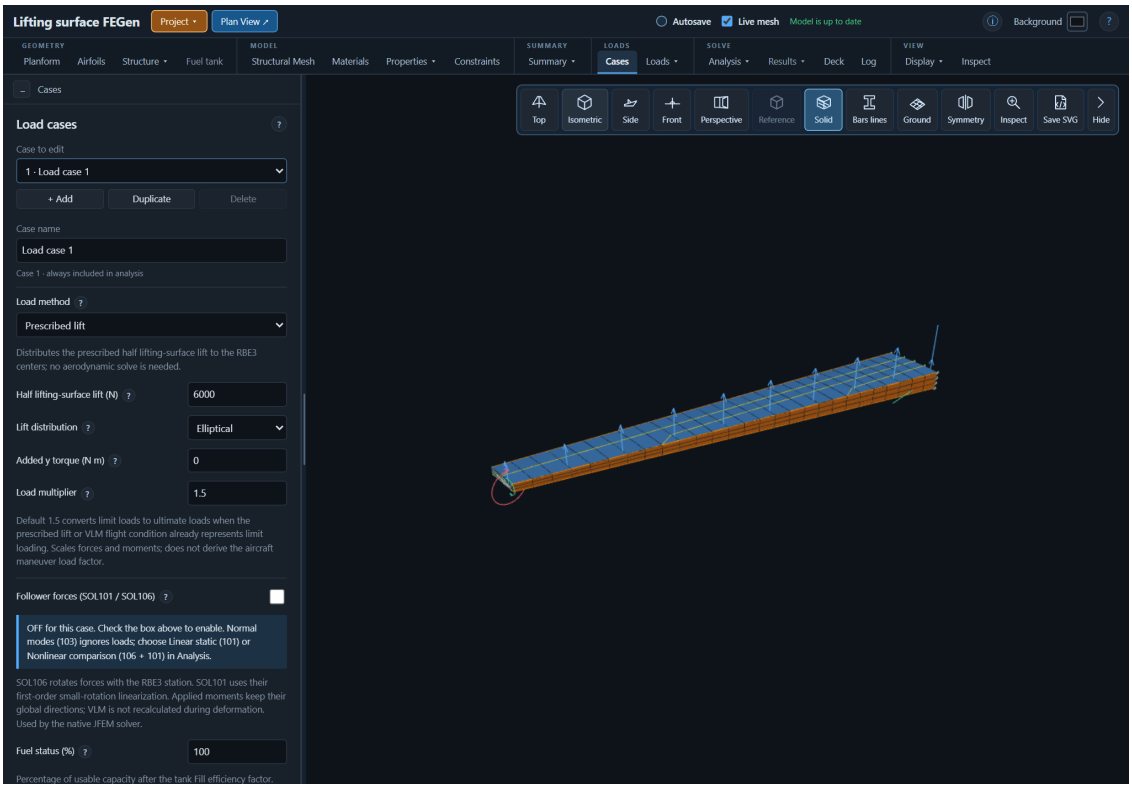
8 Load cases and aerodynamic loads

Cases is the starting point for loading. Give each case a useful name, enable the cases to run, and review the multiplier and case-specific inputs. Switching the active case changes what the viewport displays. The solver run processes enabled cases; the visible case alone does not define the complete run list.

Check the load distribution, load direction, resultant and application points. Generated cases can combine aerodynamic and inertial effects, including fuel where enabled. A change in mass or fuel state can therefore change the preload even if the visible aerodynamic distribution looks similar.

Loads > VLM settings controls the aerodynamic discretization and calculation settings. **Load plots** helps check spanwise trends and resultants. Use an aerodynamic mesh study when the aerodynamic load is sensitive to discretization, and a structural mesh study for FE response. These are different checks.

Display > Loads&Constraints only changes visibility: forces, moments, VLM loads/pressures, connection/constraint symbols and recovered reactions. The **RNE3 for Loads** switch identifies the displayed load-transfer spiders; it does not change the deck's connection definition. A hidden arrow is still a load in the model.



Review the enabled case list and each case's loading before Run in JFEM. Keep one enabled case for a first tutorial run.

Record load assumptions in Study comments. When comparing two runs, hold the physical case and boundary conditions fixed unless the intended study is specifically about those changes.

9 Choose an analysis

Under **Analysis > FE Analysis**, select the solution and its controls. Check the number of requested modes/roots, search interval where offered, output directory and timeout. Then click **Run in JFEM**. The application prepares the deck, runs the solver and reads the results; these are distinct phases.

Solution	Question	Interpretation
SOL101	Linear static response	Displacement, stress and reactions for the specified load state. Verify equilibrium, constraint assumptions and linear applicability.
SOL103	Free vibration modes	Frequencies are in Hz for consistent physical input. Mode amplitudes are normalized, not a forced-vibration displacement prediction.
SOL105	Linear buckling	Roots are dimensionless factors on the static preload. Modes indicate an instability pattern; amplitude is arbitrary.
SOL106	Geometrically nonlinear static response	Follow the accepted load steps and convergence. This path is experimental; review the paired linear comparison and solver diagnostics.

Modal extraction and EIGRL

For generated SOL103, the EIGRL frequency bounds are left blank and mass normalization is requested. The requested mode count limits the extraction. For generated SOL105, the lower bound is 10^{-6} and the upper limit is the selected buckling search factor. This is a factor range, not Hz. The help shows the actual exported values.

The compact fixed-width card `EIGRL 11.0000-650.00000 10` has fields `EIGRL | 1 | 1.0000-6 | 50.00000 | 10`: SID 1, lower bound 10^{-6}, upper bound 50 and ten requested roots. The apparent run-together numbers are fixed-field formatting, not one number. Imported decks retain their actual source extraction settings until explicitly overridden.

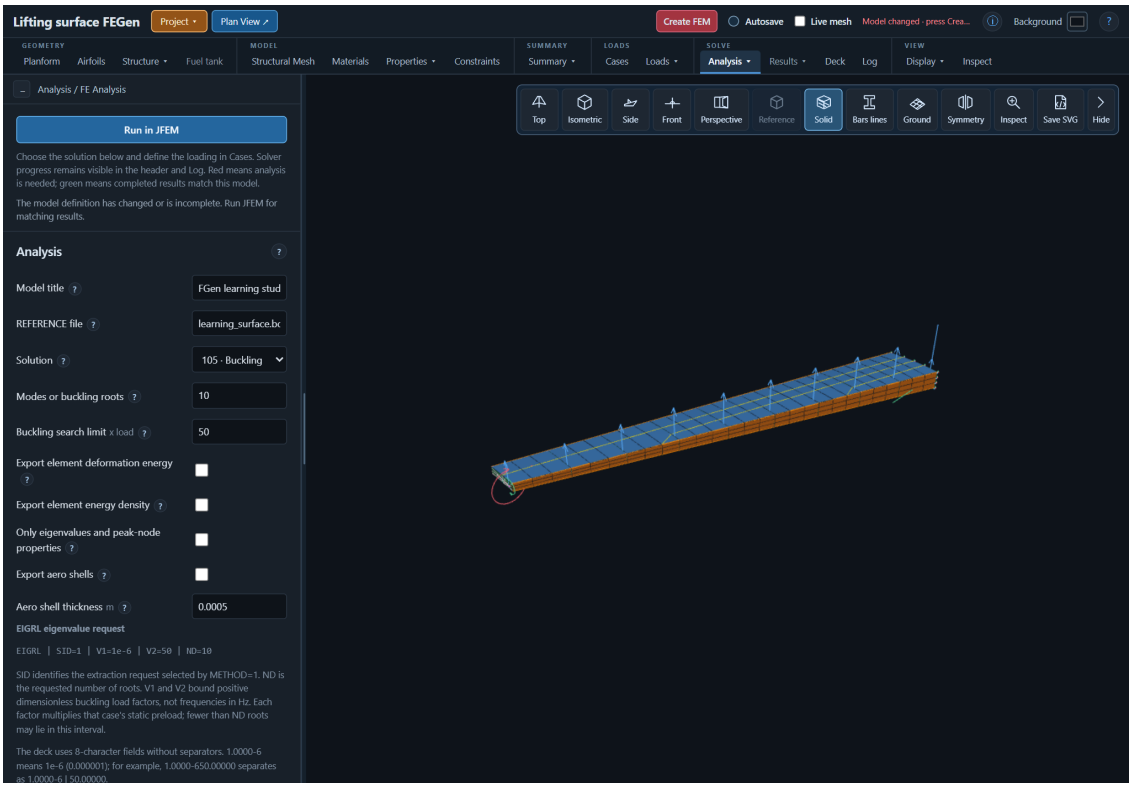
Nonlinear progress

For SOL106, the load clock distinguishes the current target from the last accepted load fraction. A run that stopped at a partial fraction is not a full-load solution. Inspect the accepted step, residual/convergence information and paired SOL101 response. Do not infer convergence from the last animation frame or from a solver process merely having ended.

10 Buckling factors and optional outputs

A SOL105 factor λ scales the complete static preload used by the eigenproblem. For a generated case with load multiplier a, the displayed base-case factor is λa. If inertia participates in that preload, it scales too. For an imported model, interpret the result against the source-deck preload; there is no additional generated-wing multiplier.

Negative roots can correspond to reversing the preload direction. Missing roots within a requested interval are not proof of safety. Check the search settings, preload, constraints and solver diagnostics, and confirm the relevant mode is a meaningful structural instability.



SOL105 extraction and output controls. Leave compact output off when deformation plots or field results are needed.

- **Export element deformation energy**: requests true elastic element energy $U_e = \frac{1}{2} u_e^T K_e u_e$ from the production element stiffness.
- **Export element energy density**: requests U_e/V_e where a physical element volume is defined. A zero-volume spring has no defined density; unavailable values are not zeros.
- **Only eigenvalues and peak-node properties**: writes `PARAM,JFBKCOMP,1`. It preserves roots, preload/subcase identities and peak-translation node/property metadata, but omits exported vectors, preload fields and energy data. It takes precedence over the two energy options.

All three default to off. Compact output does not change the eigenproblem or its eigenvalues. Eigenvectors are still calculated internally. It reduces export/recovery work and file size; it is not a smaller eigenproblem or an accuracy control. Disable compact output and rerun to obtain missing graphical results.

11 Read buckling metadata correctly

The peak-node report uses the Euclidean translational magnitude, excluding rotations. It retains ties within a relative tolerance of 10^{-6}. Property IDs are those of elements directly incident on each peak GRID. One GRID may touch several properties or none. In particular, a reference GRID of a rigid load-transfer connection is not assigned a nearby shell's PID merely because the shell is geometrically close.

Use the PID report to locate a mode, then inspect the mode shape and model connectivity when full output is available. It does not identify a failed material, calculate a reserve factor for each property, or replace interpretation of the buckling deformation.

Preload energy versus modal energy

The static preload has a physical displacement amplitude determined by its applied loads. A buckling eigenvector has an arbitrary normalization. Its element energies scale with the square of the chosen eigenvector amplitude. Therefore modal energy is useful for locating deformation within that normalized mode, but is not the stored energy at the critical physical load.

Exports separate the static preload and each mode. They state normalization, coverage, exclusions and units. The elastic energy excludes geometric stiffness and constraint penalties; element numerical stabilization is included in the production elastic operator. Unsupported contributions remain explicitly unavailable. Thermal preload stored energy may be unavailable when the mechanical-energy correction is unsupported.

Generated N–m models report energy in J and density in J/m^3. A Reference deck has no universal declared unit system. Energy follows force times length; density follows force divided by length squared; volume follows length cubed. Do not relabel an N–mm deck as N–m without a complete unit conversion.

Imported output requests

An imported SOL105 source retains its output requests. Enable **Override source SOL105 output settings** only when you intend to replace them. The Study saves this overlay. `JFBKENG`, `JFBKDEN` and `JFBKCOMP` are JFEM-specific optional export parameters, not material or stiffness properties.

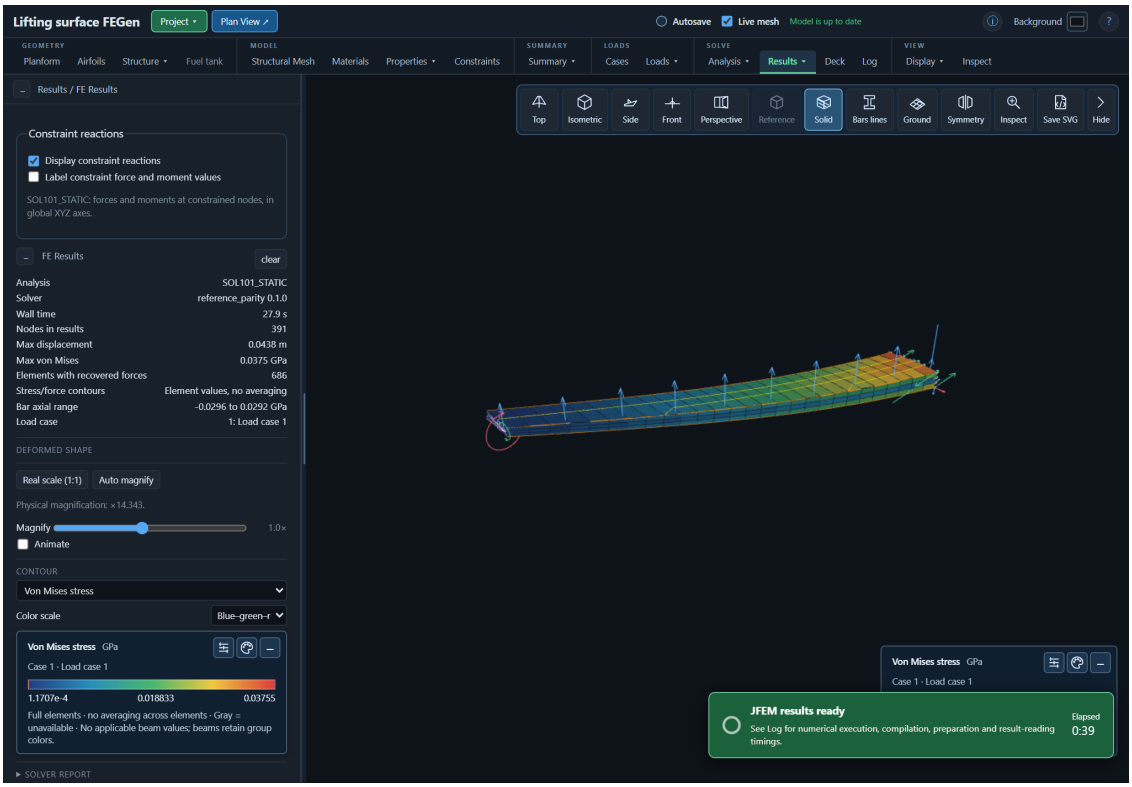
12 Explore FE Results

Open **Results > FE Results** after a completed run. Select the load case, analysis and field before interpreting the colors. For a normal SOL105 run, **SOL101 preload** shows the physical static state used in buckling, while **SOL105 buckling** shows the normalized modes. Older or compact outputs may not contain all fields.

Use deformation scale and animation as viewing aids. Read the numerical displacement and scale instead of estimating deflection from the picture. For modal and buckling plots, amplitude is normalized. The undeformed reference helps retain the physical geometry.

Stress legends, ranges and inspection tables display GPa; exported/stored values retain their explicit canonical units. Table values use at most five significant digits for readability; rounding and colors do not change stored results or CSV values. Review the selected component and frame. Set legend limits deliberately when comparing cases: automatic ranges can make different magnitudes look similar.

Display constraint reactions directly toggles the recovered reaction vectors. Enabling it reveals the independent **Label constraint force and moment values** control. These settings are synchronized with **Display > Loads&Constraints**. Values only exist when recovered for the selected result; showing symbols cannot create missing data.



Real SOL101 results for the synthetic tutorial model. The numerical values apply only to this example and its default assumptions.

Definition changes can make a retained result stale. Check the result status and rerun when needed. A saved Study alone does not embed every solver field file. Keep its corresponding results directory.

13 Sensitivity: a controlled local study

Analysis > Sensitivity evaluates how a selected response changes with model variables. First obtain and understand an appropriate baseline analysis. Select the solution/case and objective, then read the model properties and select the variables to differentiate. Selecting or refreshing variables does not launch a calculation.

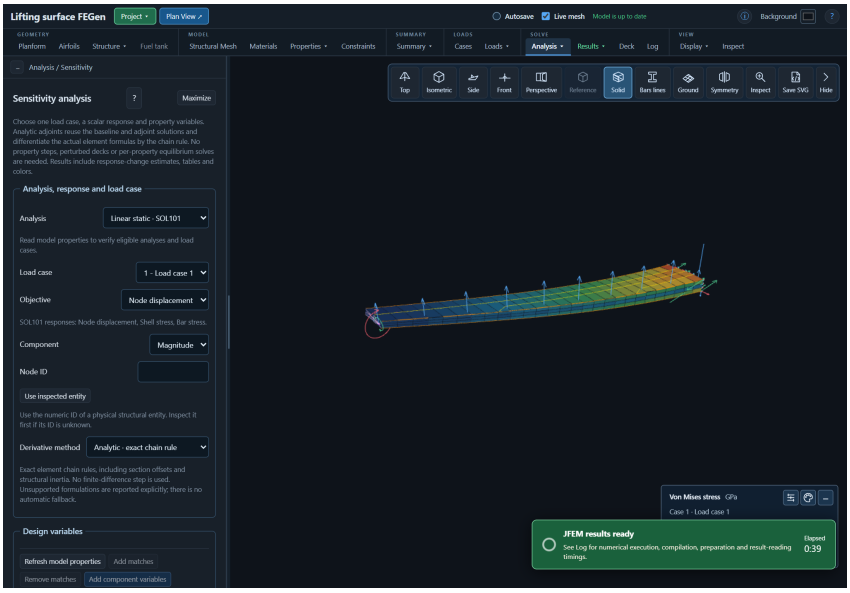
Component-dimension selections expand into local bay/rib variables, including applicable overrides. Review their locations and IDs. **Add component variables** selects all eligible local component dimensions; start with a small relevant set. Material variables retain their shared scope. Imported models offer only native variables accepted by the method's formulation checks.

Use **Wildcards** (the default) for patterns: `P*` or the prefix shortcut `P_` narrows property names, `*skin*` contains "skin", and `*bar*lower*` requires those terms in order. **Text** instead matches a literal substring. Choose **Regex** explicitly for a regular expression. **Add matches** is additive: existing selections stay selected.

Objectives depend on solution: SOL101 offers node displacement, shell stress and bar stress; SOL103 offers a modal eigenvalue; SOL105 offers a buckling factor; SOL106 offers final-state displacement. Native capability checks govern availability for the actual formulation and model. SOL106 requires the analytic final-state adjoint and an accepted nonlinear baseline; a dropdown choice alone does not establish method support.

Click **Run sensitivity** after reviewing the method and its applicability. The activity panel can move between baseline calculation, operator construction and adjoint stages. Progress measurements without a compilation split are reported as wall time, not silently treated as pure execution.

In **Sensitivity results**, inspect the objective, variable, derivative units and status together. A property response map repeats a property derivative over its associated elements; it is not a local strain-energy contour. Derivative sign and magnitude describe the stated objective and perturbation convention. They do not by themselves establish a feasible redesign.



Sensitivity setup following the baseline static calculation. Select and review variables before starting the separate sensitivity run.

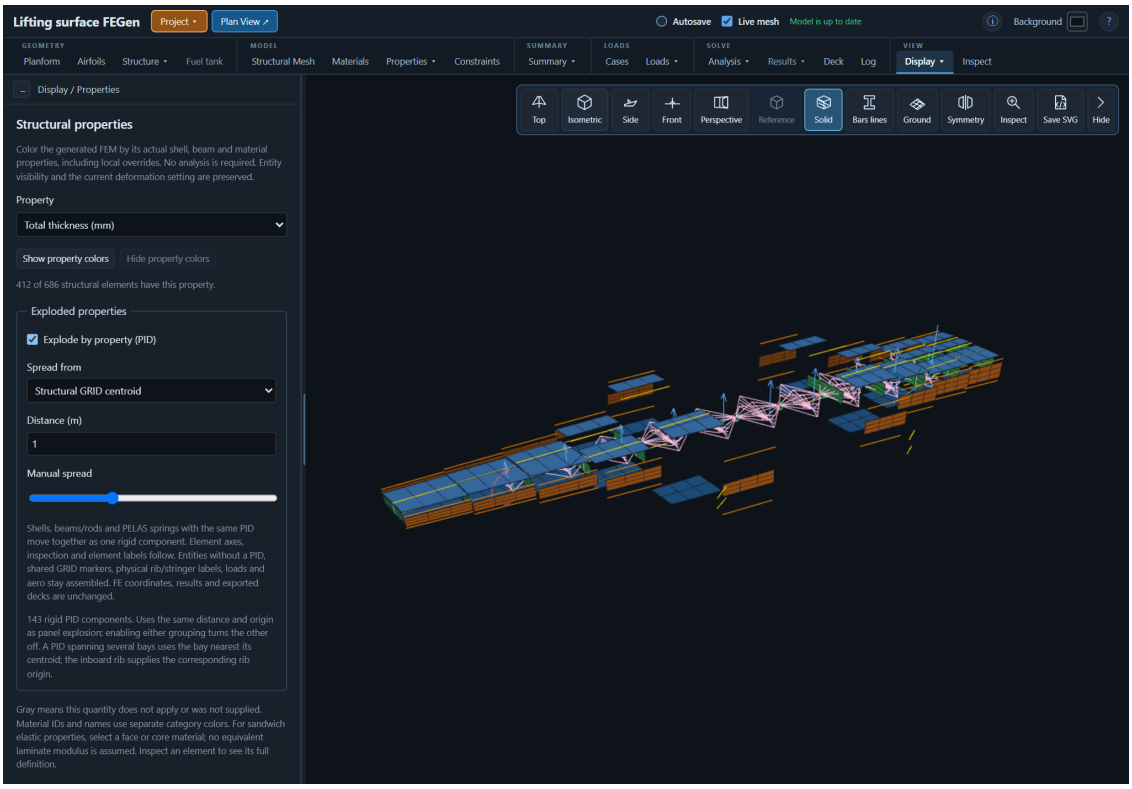
Missing, pending, failed and not-selected entries are different states. Older shared-component derivatives are not divided or copied into new local variables. Rerun when independent local derivatives are required. Check promising changes with a fresh forward solve.

14 Display, inspection and property explosion

The **Display** menu changes the view. It does not change loads, materials or geometry. Use its subgroups to find the relevant controls: entities, geometry, loads and constraints, properties, appearance, axes and labels, environment, and mass properties.

The viewport toolbar provides Top, Isometric, Side, Front and projection controls. Double-click empty canvas to fit while retaining the current orientation. Use **Inspect**, choose an entity type and click a matching visible entity to see its data. Click Inspect again to leave inspection mode. Hide unrelated entities to make picking and connectivity checks easier.

Display > Properties > Explode by property (PID) moves every element with the same actual PID together. **Explode panels** groups the skin and stringers of a stiffened panel instead. The groupings are mutually exclusive and share the distance/origin settings. Origins include the structural GRID centroid, bounding-box center and, where metadata exists, rib-bay or inboard-rib centers.



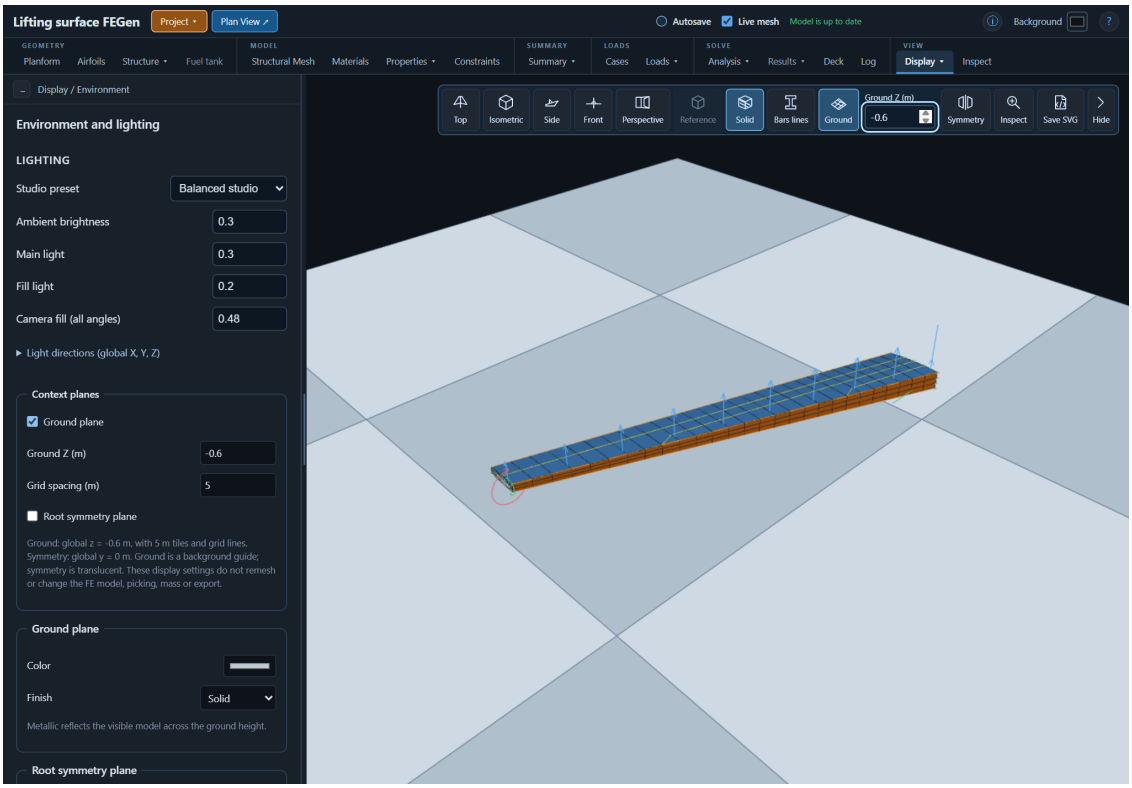
Property explosion separates actual property groups for visual inspection. It does not modify physical node coordinates.

Moved distances are display distances. Loads, common node markers and other unmoved references need not follow the exploded parts. Results keep their physical values; do not measure a physical gap from an exploded picture. Set the distance to zero or disable explosion to restore the assembled view. SVG captures the current view, whereas physical STL/GLB exports retain assembled geometry. Imported models without rib metadata cannot use rib-based origins.

15 Environment, transparency and reactions

Display > Environment contains the ground and symmetry planes. Adjust ground elevation in global Z, grid spacing, color, finish and opacity. These are display aids, not contact surfaces or boundary conditions. The ground toolbar shortcut and Environment controls refer to the same plane.

Wireframe shows the grid; translucent adds a transparent plane; metallic includes a reflection of visible model entities. Ground spacing and extent are shared across finishes. Use opacity and the structural surface finish together to keep important internal components visible. A reflection is a visual aid, not a second modeled structure.



Environment collects the ground and symmetry settings. Their visibility, appearance and elevation are saved with the Study.

Display > Loads&Constraints collects applied force/moment arrows, aerodynamic loads/pressures, connection/constraint symbols and recovered reactions. After selecting a result containing reactions, enable **Display constraint reactions**; then enable value labels only when needed. The same two controls are available in FE Results and stay synchronized.

If a view becomes cluttered, switch off labels first, then unrelated entities. If an expected object is absent, check its group visibility, selected case, result field and availability before changing the model definition. A display toggle never adds a missing load, reaction or FE entity.

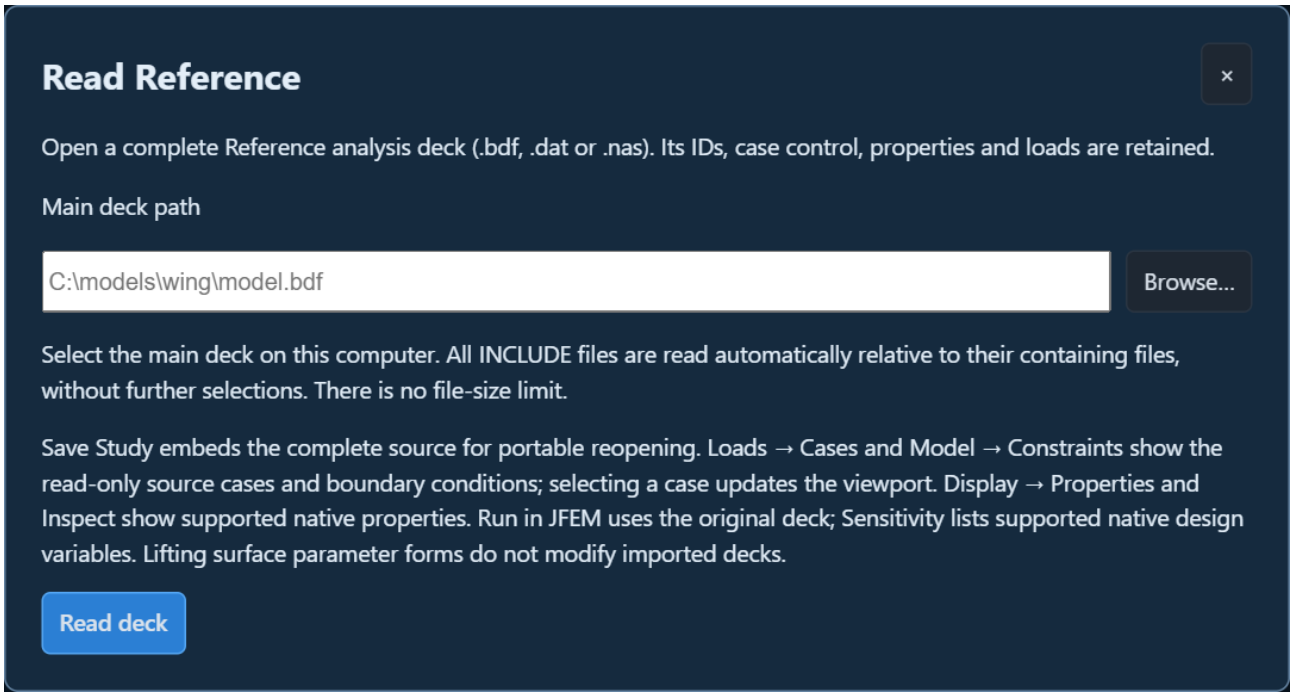
16 Read Reference and inspect the deck

Project > Read Reference opens an existing `.bdf`, `.dat` or `.nas`. Browse to the main deck or paste its complete local path. INCLUDE files are resolved relative to their containing files. Wait for importer preparation and the read operation to finish; file reading and geometry construction are separate reported stages.

The original deck is authoritative. IDs, case control, properties, loads and source constraints are retained. Generated lifting-surface geometry controls cannot reshape an imported deck. **Cases** and **Constraints** expose its source definitions as read-only information; case selection updates the view. Supported native properties appear in Properties and Inspect.

Analysis > FE Analysis > Imported deck analysis can retain the source solution or apply an explicit supported analysis overlay. Check the actual extraction and output settings. Preserve the source settings unless a change is intentional; save the overlay with the Study. Sensitivity lists supported native variables rather than pretending the deck is a generated wing.

Saving an imported Study embeds the source and INCLUDE contents for portable reopening. Keep the original source as well when maintaining a deck-based engineering workflow. The Plan View button is disabled when generated geometry metadata is unavailable.



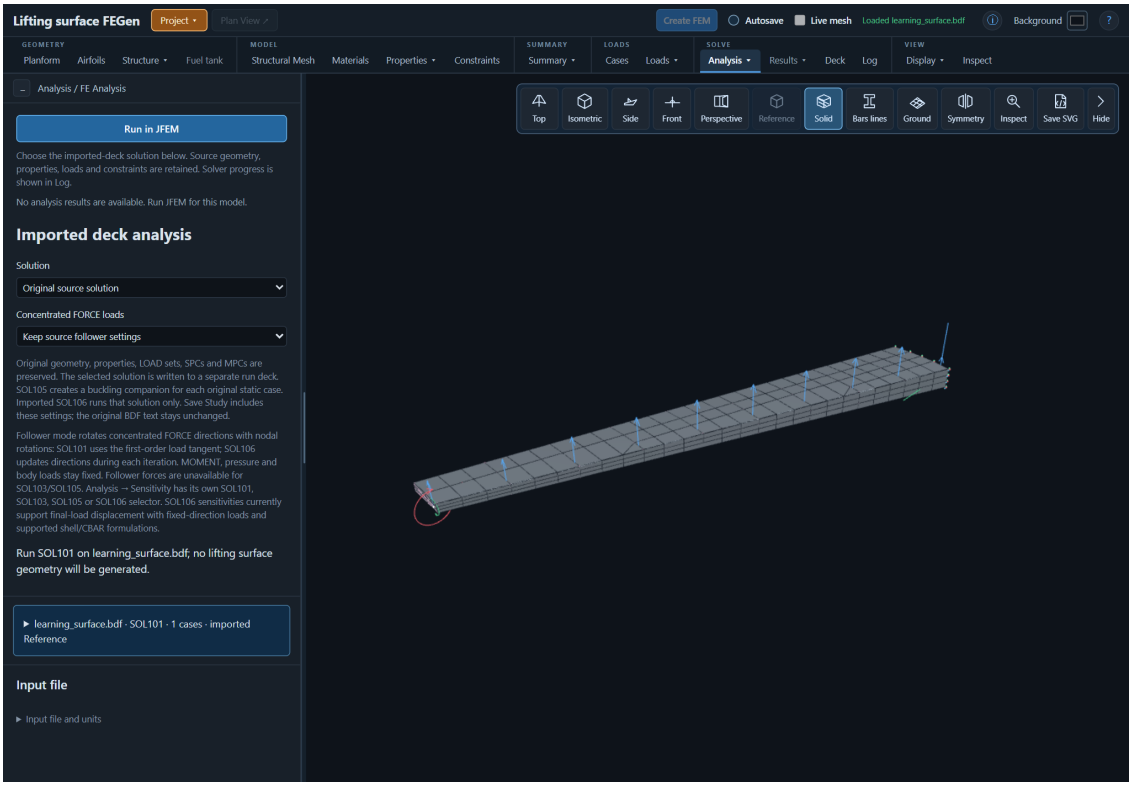
Read Reference accepts a local main-deck path and resolves its INCLUDE tree. This dialog is from the same isolated tutorial session.

17 Deck review and an imported-model example

Project > Write deck prepares the generated deck for inspection/export. Open **Deck**, search for a card or ID, select/copy text, or use the available download controls. Large decks retain complete text; the viewer optimizes layout without deleting cards. Reopening reuses the loaded deck until a new Write/Run output replaces it.

Start at executive/case control: check SOL, SUBCASE, SPC, LOAD and METHOD/EIGRL where applicable. Then check representative GRID, element, property and material cards. The generated deck documents its consistent units and coordinate convention. In an imported deck, use its own documented units.

For the example below, the tutorial's freshly generated SOL101 deck was read back through Read Reference. Its FE topology is retained, while the generated planform editor is no longer authoritative. The source analysis remains selected until an overlay is requested. This is a useful learning exercise: write a generated deck, read that same deck, and compare the retained nodes, properties and cases.



The tutorial deck reopened as an imported Reference model, using the actual native importer. Its source analysis and properties are retained.

The Deck viewer contains a saved snapshot. Later form edits do not rewrite that text until Write deck or Run in JFEM creates a new output. Use the snapshot's metadata to identify it.

18 Activity, elapsed time and the detailed Log

The bottom-right panel names the current task. Its clock is total task *elapsed* time, including preparation, possible Julia compilation and file operations. It is not a pure solver stopwatch.

Color	Meaning
Orange	Startup/preparation, explicitly reported compilation or precompilation, importing, export and result/file reading.
Blue	A reported numerical solver phase. Julia may also compile during this phase; the live label says so when compilation is inseparable.
Green	The task completed successfully. Review result availability and engineering convergence separately.
Red	An actual task error. Read the error and detailed Log before retrying.

Julia compilation counters are retrospective. Blue identifies the numerical phase; it does not assert that every moment of its live elapsed time was free of compilation. Cold startup and first-use code can take longer than a later run. Do not infer a measured precompile duration from all startup time.

When a run finishes, **Log** reports available category totals and detailed stage rows: stage name, call count, wall time, compilation/recompilation, execution and garbage collection. Recompilation is included in compilation; GC is already included in wall time. A parent's inclusive wall time already contains its children. Add exclusive contributions or use category totals; do not add both parent and child inclusive rows.

Sensitivity operator/adjoint measurements that lack a reliable compilation split remain explicitly unseparated wall measurements. Missing measurements are not zeros. Result loading can continue after numerical work ends, so its activity correctly returns to orange until the results are ready.

For a useful comparison, record: the model/revision, active solution and cases, mesh size, requested outputs, cold or warm process context, total elapsed time, measured numerical execution and compilation where available. Compact buckling output can reduce recovery/export cost without changing the eigenproblem. Compare like-for-like output requests.

19 Troubleshooting and a review checklist

Symptom	Next check
The page cannot connect	Keep the local server running. Use its reported address and port. Confirm the browser is opening that instance.
First operation is slow	Read the current phase and Log. Package loading and first-use compilation can dominate a cold run; total elapsed is not pure solve time.
Model or results are stale	Rebuild after definition changes, then rerun the intended cases. View-only changes do not require a solve.
No result contour after SOL105	Check whether compact output was requested, whether vectors were exported, and which analysis/field is selected. Disable compact output and rerun for vectors.
Singularity or implausible motion	Review constraints, disconnected regions, material/section units and load directions. Do not add arbitrary constraints solely to suppress an error.
SOL106 ends at partial load	Read the last accepted fraction and convergence diagnostics. Partial progress is not a full-load result.
Missing imported INCLUDE	Verify the main path and referenced relative files. Read the actual error; do not substitute an unrelated file.
Missing reaction arrows	Select a result with recovered reactions and enable their display. Labels are a separate toggle.
Wrong-looking scale or color	Check physical units, chosen component/frame, deformation multiplier and manual legend limits.
Saving created another file	Check direct-write browser support and download behavior. Keep the intended Study revision and separate solver outputs.

Before relying on a result

1. Confirm the model definition, units, materials, effective properties and mass.
2. Confirm connectivity, constraints, selected load case and full preload interpretation.
3. Confirm the solution, requested fields, extraction interval and convergence/status.
4. Check equilibrium and expected trends; perform an appropriate mesh study.
5. Distinguish physical static fields from normalized modes and sensitivity derivatives.
6. Save the Study with a useful comment and retain the matching deck/results folder.

Scope of this guide. Screenshots document the October 2026 interface using a synthetic default model. The guide is an application workflow reference. Solver formulations, supported elements, validation cases and installation requirements are documented separately in the OpenJFEM repository and its reference documentation.